

Quantum Information Theory

Peter Zeman

NMMB537

Preface

These are lecture notes for the course NMMB537 Quantum Information Theory, taught at Faculty of Mathematics and Physics, Charles University. The lecture notes are mostly inspired by the course NMAK14020U Quantum Information Theory, offered at University of Copenhagen.

Contents

Preface	i
1 Quantum States and Measurements	1
1.1 Finite-dimensional Hilbert Spaces	1
1.2 Quantum States	4
1.3 Bloch Sphere	5
1.4 Measurements	7
1.5 Bloch Sphere Revisited	8
1.6 Observables	9

Chapter 1

Quantum States and Measurements

We start by describing a precise mathematical framework for quantum states.

1.1 Finite-dimensional Hilbert Spaces

Axiom 1 (Hilbert space). To every quantum system we associate a Hilbert space $\mathcal{H}$.

We only consider finite-dimensional Hilbert spaces in this course. If $\dim(\mathcal{H}) = d$, then $\mathcal{H}$ is isomorphic to

$$\mathbb{C}^d = \left\{ \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_d \end{pmatrix} : \psi_i \in \mathbb{C} \right\}$$

with the standard inner product

$$\langle \psi | \phi \rangle = \sum_{i=1}^d \overline{\psi_i} \phi_i.$$

Bra-ket notation. We write $|\psi\rangle \in \mathcal{H}$ for any vector. By $\langle\psi| \in \mathcal{H}^\dagger$, we mean the linear map $\mathcal{H} \rightarrow \mathbb{C}$ defined by $\langle\psi|(|\phi\rangle) = \langle\psi | \phi\rangle$. In other words, $\langle\psi|$ is the row vector corresponding to the conjugate transpose of $|\psi\rangle$. We denote the standard basis of $\mathbb{C}^d$ by $|0\rangle, |1\rangle, \dots, |d-1\rangle$. With coordinates numbered $1, \dots, d$, the vector $|i\rangle$ has 1 in coordinate $i+1$ and 0 elsewhere.

Linear algebra notation. Every basis of a Hilbert space that we consider will be orthonormal. If there is no confusion, we will use the terms linear map and matrix interchangeably.

By $\text{Lin}(\mathcal{H}, \mathcal{K})$, we denote the set of linear maps $\mathcal{H} \rightarrow \mathcal{K}$. When we fix the standard bases, we can express $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$ as

$$M = \sum_{i,j} M_{ij} |i\rangle\langle j|.$$

Note that $\langle i|M|j\rangle = M_{ij} \in \mathbb{C}$.

Exercise 1.1. Write $\sum_{i,j \in \{0,1\}} ij|i\rangle\langle j|$ as a matrix.

If $M \in \text{Lin}(\mathcal{H}, \mathcal{H}) = \text{Lin}(\mathcal{H})$, then the trace with respect to the standard basis is

$$\text{tr}[M] = \sum_i M_{ii} = \sum_i \langle i|M|i\rangle.$$

An important fact about trace is that it is in fact independent of the basis. Recall also that $\text{tr}[MN] = \text{tr}[NM]$.

Here are some important types of matrices:

- $M \in \text{Lin}(\mathcal{H})$ is *Hermitian* if $M^\dagger = M$.
- $P \in \text{Lin}(\mathcal{H})$ is *positive* if $\langle \psi | P | \psi \rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$. $\text{PSD}(\mathcal{H})$ is the class of all positive matrices. We also use the notation $P \geq 0$ to indicate that P is positive. If $M, N \in \text{Lin}(\mathcal{H})$, then we define $M \geq N$ if $M - N \geq 0$, that is, if $M - N$ is positive.
- $U \in \text{Lin}(\mathcal{H})$ is *unitary* if $U^\dagger U = U U^\dagger = I$. $\mathcal{U}(\mathcal{H})$ is the class of all unitary matrices.
- $V \in \text{Lin}(\mathcal{H}, \mathcal{K})$ is an *isometry* if $V^\dagger V = I$.
- $P \in \text{Lin}(\mathcal{H})$ is a *projection* if $P^\dagger = P$ and $P^2 = P$.

Exercise 1.2 (Why isometries preserve geometry). Show that $V \in \text{Lin}(\mathcal{H}, \mathcal{K})$ satisfies $V^\dagger V = I$ if and only if it preserves inner products. Deduce that it preserves lengths and distances.

Solution. For any $|\psi\rangle, |\phi\rangle \in \mathcal{H}$,

$$\langle V\psi | V\phi \rangle = \langle \psi | V^\dagger V | \phi \rangle.$$

If $V^\dagger V = I$, this equals $\langle \psi | \phi \rangle$. Conversely, if inner products are preserved, then $\langle \psi | (V^\dagger V - I) | \phi \rangle = 0$ for every pair of vectors. Taking basis vectors shows that every matrix entry of $V^\dagger V - I$ is zero, so $V^\dagger V = I$.

Taking $|\phi\rangle = |\psi\rangle$ gives $\|V|\psi\rangle\|^2 = \||\psi\rangle\|^2$. By linearity, this also gives

$$\|V|\psi\rangle - V|\phi\rangle\| = \|V(|\psi\rangle - |\phi\rangle)\| = \||\psi\rangle - |\phi\rangle\|.$$

For example, $V : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ given by $V(a, b) = (a, b, 0)$ is an isometry. Its image can be a proper subspace of the target space. $\square$

Exercise 1.3 (Why these operators are projections). Show that the conditions $P^2 = P$ and $P^\dagger = P$ mean that P is the orthogonal projection onto its image $W = \text{ran}(P)$.

Solution. If $|w\rangle \in W$, then $|w\rangle = P|u\rangle$ for some $|u\rangle$, so $P|w\rangle = P^2|u\rangle = P|u\rangle = |w\rangle$. Thus P leaves every vector in its image fixed. For an arbitrary vector, write

$$|\psi\rangle = P|\psi\rangle + (I - P)|\psi\rangle.$$

The first term lies in W . To see that the second is perpendicular to W , take any $|w\rangle \in W$ and use $P|w\rangle = |w\rangle$, $P^\dagger = P$, and $P(I - P) = 0$:

$$\langle w | (I - P) | \psi \rangle = \langle Pw | (I - P) | \psi \rangle = \langle w | P(I - P) | \psi \rangle = 0.$$

Hence P keeps the component in W and removes the component in $W^\perp$. Conversely, an orthogonal projection has matrix $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ in an orthonormal basis adapted to W , where $r = \dim W$. This matrix satisfies both conditions.

For example,

$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} (a+b)/2 \\ (a+b)/2 \end{pmatrix}$$

is the orthogonal projection onto the line spanned by $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. □

Exercise 1.4. Are the following matrices {Hermitian, PSD, unitary, projections}?

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}.$$

Theorem 1.5 (Spectral theorem for Hermitian matrices). *If $M \in \text{Lin}(\mathcal{H})$ is Hermitian $d \times d$ matrix, then M has eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$ and there is an orthonormal basis of eigenvectors $\{|\psi_i\rangle\}_{i=1}^d$ such that*

$$M = \sum_{i=1}^d \lambda_i |\psi_i\rangle\langle\psi_i|.$$

Recall that the eigenvalues can have multiple occurrences and the basis of eigenvectors may not be unique. Indeed, for instance,

$$I = \sum_{i=1}^d |\psi_i\rangle\langle\psi_i|$$

is true for any orthonormal basis $\{|\psi_i\rangle\}_{i=1}^d$.

This allows us to define functions of Hermitian matrices. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function, then we can define

$$f(M) = \sum_{i=1}^d f(\lambda_i) |\psi_i\rangle\langle\psi_i|.$$

For instance, if $M \geq 0$, the same construction with $f(t) = \sqrt{t}$ on $[0, \infty)$ defines its positive square root:

$$\sqrt{M} = \sum_{i=1}^d \sqrt{\lambda_i} |\psi_i\rangle\langle\psi_i|.$$

Note that if P is a projection, then the condition $P^2 = P$ implies that the eigenvalues are in $\{0, 1\}$. Thus, we can write $P = \sum_{i=1}^r |\psi_i\rangle\langle\psi_i|$. Here, $r = \text{rank}(P)$ of the projection P .

Theorem 1.6 (Characterization of positive matrices). *If $P \in \text{Lin}(\mathcal{H})$, then the following are equivalent:*

- (a) P is positive, i.e., $\langle\psi|P|\psi\rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$.
- (b) $P^\dagger = P$ and all eigenvalues are nonnegative.
- (c) There is $M \in \text{Lin}(\mathcal{H})$ such that $P = M^\dagger M$.
- (d) $\text{tr}[PQ] \geq 0$, for every $Q \in \text{PSD}(\mathcal{H})$.

Exercise 1.7. Prove that if $P \in \text{PSD}(\mathcal{H})$ and $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$, then $MPM^\dagger$ is positive.

1.2 Quantum States

Definition 1.8. A *density matrix* is a positive operator $\rho \in \text{PSD}(\mathcal{H})$ with $\text{tr}[\rho] = 1$. We put $\mathcal{S}(\mathcal{H}) = \{\rho \in \text{PSD}(\mathcal{H}) : \text{tr}[\rho] = 1\}$.

Axiom 2. The state of a quantum system with Hilbert space $\mathcal{H}$ is described by a density matrix ρ . We will refer to ρ as a quantum state.

Example 1.9 (classical states). In probability theory we have a finite set of outcomes Σ with a probability distribution p such that $p(x) \geq 0$, for $x \in \Sigma$, and $\sum_{x \in \Sigma} p(x) = 1$. If $\mathcal{H}$ has an orthonormal basis $\{|x\rangle\}_{x \in \Sigma}$, then

$$\rho = \sum_{x \in \Sigma} p(x) |x\rangle\langle x|$$

is a density matrix. Conversely, any diagonal density matrix corresponds to a probability distribution.

Example 1.10 (pure states). Let $|\psi\rangle \in \mathcal{H}$ be a unit vector. Then $\rho = |\psi\rangle\langle\psi|$ is a density matrix since $\text{tr}[|\psi\rangle\langle\psi|] = \text{tr}[\langle\psi|\psi\rangle] = 1$. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$|\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\bar{\beta} \\ \bar{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

Note that replacing $|\psi\rangle$ by $e^{i\theta}|\psi\rangle$ does not change the density matrix.

Exercise 1.11. Is $\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ a pure state?

Solution. Yes, since $\rho = |+\rangle\langle+|$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. □

Example 1.12 (general state). Let ρ be a general quantum state. By Theorem 1.6, we can write

$$\rho = \sum_{i=1}^d p_i |\psi_i\rangle\langle\psi_i|,$$

where $p_i \geq 0$. Since ρ is diagonal in the basis $\{|\psi_i\rangle\}_{i=1}^d$ and trace is independent of the chosen basis, $\text{tr}[\rho] = 1$ implies that $\sum_{i=1}^d p_i = 1$. So, the eigenvalues p_i 's define a probability distribution. We may interpret ρ as if we have the pure state $|\psi_i\rangle\langle\psi_i|$ with probability p_i . We call ρ a *mixed state* if it is not pure.

If the spectrum is nondegenerate, this spectral decomposition is unique up to ordering and phases of the eigenvectors. Nevertheless, a mixed state admits other *ensemble decompositions* into pure states that need not be orthogonal, with different probabilities and possibly different numbers of states.

Exercise 1.13. What is the state corresponding to 50% chance of $|0\rangle$ and 50% chance of $|+\rangle$?

Solution.

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Note that the eigenvalues of ρ are $1/2 + \sqrt{2}/4$ and $1/2 - \sqrt{2}/4$, so the spectral decomposition of ρ gives a completely different decomposition of ρ compared to the one we started with. $\square$

Definition 1.14. The state $\tau = \frac{1}{d}I$ is called the *maximally mixed state*.

Exercise 1.15 (Different preparations of the same state). Compare the state obtained by choosing $|0\rangle$ or $|1\rangle$ with equal probabilities with the state obtained by choosing $|+\rangle$ or $|-\rangle$ with equal probabilities, where $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Solution. The first preparation gives

$$\rho_Z = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

For the second preparation, the off-diagonal entries cancel:

$$\rho_X = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{I}{2}.$$

Thus both preparations give the same maximally mixed state. In contrast, preparing the superposition $|+\rangle$ every time gives $|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Adding state vectors to form a superposition and averaging density matrices to describe a random preparation are different operations. $\square$

1.3 Bloch Sphere

How does the set $\mathcal{S}(\mathcal{H})$ look like? For two states $\rho_1, \rho_2 \in \mathcal{S}(\mathcal{H})$ and $t \in [0, 1]$, the convex combination $t\rho_1 + (1-t)\rho_2$ is also a state. In fact, we have the following lemma.

Lemma 1.16. *The set $\mathcal{S}(\mathcal{H}) \subseteq \text{Lin}(\mathcal{H})$ is convex. Moreover, the extreme points of $\mathcal{S}(\mathcal{H})$ are exactly the pure states.*

For the particular case of one qubit, there is a visualization known as the Bloch sphere. Hermitian 2×2 matrices have the following (real) basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The last three matrices are known as Pauli matrices. Here are some of their properties:

$$X^2 = Y^2 = Z^2 = I, \quad XY = iZ = -YX, \quad \text{tr}[X] = \text{tr}[Y] = \text{tr}[Z] = 0.$$

In particular, Pauli matrices are unitary and have eigenvalues ± 1 .

Every 2×2 Hermitian matrix can be written in a unique way as a linear combination of the matrices I, X, Y, Z with all coefficients being real numbers. An arbitrary 2×2 Hermitian matrix ρ with $\text{tr}[\rho] = 1$ can be expressed as

$$\rho = \frac{1}{2}(I + xX + yY + zZ) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}. \quad (1.3.1)$$

The coefficient $1/2$ in front of the I is fixed by the condition $\text{tr}[\rho] = 1$. Further, $\text{tr}[\rho] = 1$ implies that the eigenvalues of ρ are λ and $1 - \lambda$, for some $\lambda \in \mathbb{R}$. If we want ρ to be a quantum state, it also has to be positive, which is equivalent to requiring that $\lambda \geq 0$ and $1 - \lambda \geq 0$. Now, λ and $1 - \lambda$ are nonnegative if and only if $\det(\rho) = \lambda(1 - \lambda) \geq 0$. We compute

$$\det(\rho) = \frac{1}{4}((1+z)(1-z) - (x+iy)(x-iy)) = \frac{1}{4}(1 - x^2 - y^2 - z^2).$$

If we let $r = (x, y, z) \in \mathbb{R}^3$, then $\rho \geq 0$ if and only if $\|r\| \leq 1$. If ρ is a pure state, then $\text{rank}(\rho) = 1$ and $\det(\rho) = 0$, which is equivalent to $\|r\| = 1$. Thus, pure states correspond exactly to the unit sphere in $\mathbb{R}^3$ and mixed states correspond to the open unit ball.

Exercise 1.17. What is the pure state $|\psi\rangle$ and its density matrix $|\psi\rangle\langle\psi|$ corresponding to $r_1 = (0, 1, 0)$ and to $r_2 = (0, -1, 0)$.

Solution. By plugging r_1 into Eq. (1.3.1), we get

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

We can easily calculate, that the eigenvector corresponding to the eigenvalue 1 of ρ_1 is $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$. By Theorem 1.5, we get $\rho_1 = |\psi_1\rangle\langle\psi_1|$. Similarly, we get that $\rho_2 = |\psi_2\rangle\langle\psi_2|$, where $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$. $\square$

We have the following important bases of pure states that correspond to the intersection points of x, y, z -axes with the Bloch sphere:

- Z -basis: $\{|0\rangle, |1\rangle\}$, $r = (0, 0, 1)$, $-r = (0, 0, -1)$.
- X -basis: $\{|+\rangle, |-\rangle\}$, $r = (1, 0, 0)$, $-r = (-1, 0, 0)$
- Y -basis: $\{|+i\rangle, |-i\rangle\}$, $r = (0, 1, 0)$, $-r = (0, -1, 0)$.

Exercise 1.18 (Reading a Bloch vector). Find the Bloch vector of the state $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13. Is it pure or mixed?

Solution. Comparing entries with Eq. (1.3.1), we obtain

$$\frac{1+z}{2} = \frac{3}{4}, \quad \frac{x-iy}{2} = \frac{1}{4}, \quad \text{so} \quad r = \left(\frac{1}{2}, 0, \frac{1}{2}\right).$$

Its length is $\|r\| = 1/\sqrt{2} < 1$, so the state is mixed. The vector is also the average of the Bloch vectors of $|0\rangle$ and $|+\rangle$:

$$r = \frac{1}{2}(0, 0, 1) + \frac{1}{2}(1, 0, 0).$$

This illustrates how a probabilistic mixture becomes a convex combination of Bloch vectors. $\square$

1.4 Measurements

Definition 1.19. A *measurement* or a *positive operator valued measure (POVM)* on a Hilbert space $\mathcal{H}$ with a finite set of outcomes Ω is a function $\mu : \Omega \rightarrow \text{PSD}(\mathcal{H})$ such that $\sum_{x \in \Omega} \mu(x) = I$. If $\mu(x)$ are projections, then the measurement is called *projective*.

Axiom 3. If we measure a state ρ using μ , then the probability of getting an outcome $x \in \Omega$ is $p(x) = \text{tr}[\mu(x)\rho]$.

Remark 1.20. In general in quantum mechanics, it is not possible to perform measurements on a state without changing it. However, for now, we will not consider what happens with a state after the measurement (we can think that it is destroyed). We only get classical information about which outcome occurred. We will discuss post-measurement states later in the course.

Exercise 1.21. Prove that if ρ is a quantum state and μ is a measurement, then $p(x) = \text{tr}[\mu(x)\rho]$ defines a probability distribution on Ω .

Solution. By Theorem 1.6(d), we have $\text{tr}[\mu(x)\rho] \geq 0$ since both $\mu(x)$ and ρ are positive. By the linearity of trace,

$$\sum_{x \in \Omega} \text{tr}[\mu(x)\rho] = \text{tr} \left[\sum_{x \in \Omega} \mu(x)\rho \right] = \text{tr}[I\rho] = 1.$$

□

Example 1.22 (basis measurement). Let $\{|\psi_i\rangle\}_{i=1}^d$ be an orthonormal basis and let $\mu(i) = |\psi_i\rangle\langle\psi_i|$ be the projection onto $|\psi_i\rangle$. We get

$$p(i) = \text{tr}[|\psi_i\rangle\langle\psi_i|\rho] = \text{tr}[\langle\psi_i|\rho|\psi_i\rangle] = \langle\psi_i|\rho|\psi_i\rangle.$$

In particular, if $\rho = |\phi\rangle\langle\phi|$ is a pure state, then we get

$$p(i) = \langle\psi_i | \phi\rangle\langle\phi | \psi_i\rangle = |\langle\psi_i | \phi\rangle|^2.$$

Exercise 1.23. What are the outcome probabilities when we measure $|0\rangle$ the X -basis?

Solution. We get $|+\rangle$ with probability $|\langle+ | 0\rangle|^2 = 1/2$ and $|-\rangle$ with probability $|\langle- | 0\rangle|^2 = 1/2$. □

Exercise 1.24 (Comparing two states by measurement). Let $\rho = |+\rangle\langle+|$ and $\tau = I/2$. Calculate the outcome probabilities for each state when measuring in the Z -basis and in the X -basis.

Solution. For a Z -basis measurement, the probabilities are the diagonal entries in the standard basis. Both states have diagonal entries $1/2, 1/2$. For the X -basis measurement, the probability of outcome $+$ is

$$\langle+|\rho|+\rangle = |\langle+ | +\rangle|^2 = 1, \quad \langle+|\tau|+\rangle = \frac{1}{2}\langle+ | +\rangle = \frac{1}{2}.$$

The probability of outcome $-$ is the complement. Altogether,

state	$p_Z(0)$	$p_Z(1)$	$p_X(+)$	$p_X(-)$
ρ	$1/2$	$1/2$	1	0
τ	$1/2$	$1/2$	$1/2$	$1/2$

The two states give the same statistics in the Z -basis, but different statistics in the X -basis. Knowing the probabilities for just one measurement need not determine the state. $\square$

Exercise 1.25 (A nonprojective measurement). Show that

$$\mu(0) = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \quad \mu(1) = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}$$

define a POVM that is not projective. What are the outcome probabilities on the state $|0\rangle$?

Solution. Both matrices are positive because they are diagonal with nonnegative entries, and $\mu(0) + \mu(1) = I$. They therefore define a POVM. However,

$$\mu(0)^2 = \begin{pmatrix} 9/16 & 0 \\ 0 & 1/16 \end{pmatrix} \neq \mu(0),$$

so the measurement is not projective. The Born rule gives

$$p(0) = \text{tr}[\mu(0)|0\rangle\langle 0|] = \langle 0|\mu(0)|0\rangle = \frac{3}{4}, \quad p(1) = \frac{1}{4}.$$

One way to obtain these measurement statistics is to measure in the Z -basis and then flip the reported outcome with probability $1/4$. $\square$

1.5 Bloch Sphere Revisited

Given $r = (x, y, z)$ with $\|r\| = 1$, the positive matrices

$$\mu(0) := \rho(r) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} \quad \text{and} \quad \mu(1) := \rho(-r) = \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix}$$

define a general qubit basis measurement. Performing this measurement on a quantum state with Bloch vector $s = (x', y', z')$, then the probability of obtaining the outcome 0 is

$$p(0) = \frac{1}{2} + \frac{1}{2}(r \cdot s) = \frac{1}{2} + \frac{1}{2}(xx' + yy' + zz').$$

Geometrically $r \cdot s$ is the projection of the Bloch vector s of a quantum state onto the axis defining the measurement.

1.6 Observables

Sometimes it is convenient to reformulate measurements in terms of observables. A projective measurement is a measurement where all measurement operators are projections, that is, we have a set of outcomes Ω and projections $\{P_x : x \in \Omega\}$ such that $\sum_{x \in \Omega} P_x = I$. Suppose that $\Omega \subseteq \mathbb{R}$. The *observable* associated to the projective measurement is an operator $O \in \text{Lin}(\mathcal{H})$ given by

$$O = \sum_{x \in \Omega} x P_x.$$

Since x are real, O is Hermitian, and, conversely, each Hermitian operator O has a spectral decomposition of this form, thereby defining a projective measurement.

Remark 1.26. In physics, observables are often the preferred way to reason about measurements. Measuring the observable O is the same as performing the projective measurement defined by the spectral decomposition of O .

Example 1.27. Since $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, the observable Z corresponds to measuring in the Z -basis with outcomes $\{\pm 1\}$. Similarly, since $X = |+\rangle\langle +| - |-\rangle\langle -|$, measuring X corresponds to measuring in the X -basis with outcomes $\{\pm 1\}$. More generally, we may measure a qubit in the basis given by the antipodal points $r = (x, y, z)$ and $-r$ on the Bloch sphere with outcomes $\{\pm 1\}$. This gives an observable

$$\begin{aligned} O(r) &= \rho(r) - \rho(-r) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix} \\ &= \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}. \end{aligned}$$

States and expectation values. In classical probability, a probability distribution p on a finite set Σ assigns an expectation to each real-valued random variable $a : \Sigma \rightarrow \mathbb{R}$:

$$\mathbb{E}_p[a] = \sum_{s \in \Sigma} p(s) a(s).$$

A quantum state plays a similar role: it assigns an expectation value to each observable. The observable specifies the measurement and its numerical outcomes, while the state determines their probabilities.

Let $O = \sum_{x \in \Omega} x P_x$ and let the state be ρ . The measurement outcome is an ordinary real-valued random variable, with $p_\rho(x) = \text{tr}[P_x \rho]$ by the Born rule. We write $\mathbb{E}_\rho[O]$ for its expectation. Taking the weighted average of the outcomes and using linearity of the trace gives

$$\begin{aligned} \mathbb{E}_\rho[O] &= \sum_{x \in \Omega} x p_\rho(x) = \sum_{x \in \Omega} x \text{tr}[P_x \rho] \\ &= \text{tr} \left[\left(\sum_{x \in \Omega} x P_x \right) \rho \right] = \text{tr}[O \rho] = \text{tr}[\rho O]. \end{aligned}$$

For a pure state $\rho = |\psi\rangle\langle\psi|$, this becomes

$$\mathbb{E}_\rho[O] = \text{tr}[O |\psi\rangle\langle\psi|] = \langle\psi|O|\psi\rangle.$$

Thus, we can view a state as an *expectation functional* $O \mapsto \mathbb{E}_\rho[O]$. Just like classical expectation, it is linear, normalized, and positive: for observables A, B and $a, b \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}_\rho[aA + bB] &= a\mathbb{E}_\rho[A] + b\mathbb{E}_\rho[B], \\ \mathbb{E}_\rho[I] &= 1, \\ A \geq 0 &\implies \mathbb{E}_\rho[A] \geq 0.\end{aligned}$$

The first two properties follow from linearity of trace and $\text{tr}[\rho] = 1$; the last follows from Theorem 1.6(d). Each expectation concerns the corresponding measurement on identically prepared systems.

Probabilities as expectations. A projection P is itself an observable with possible outcomes in $\{0, 1\}$. It describes a yes-or-no question, and its expectation is the probability of the answer 1. In this sense, projections play the role of indicator random variables. In particular, the probability of outcome x when measuring O is

$$p_\rho(x) = \text{tr}[\rho P_x] = \mathbb{E}_\rho[P_x].$$

The expectation functional therefore determines the entire outcome distribution of every projective measurement. More generally, the probability of outcome x for any POVM μ is $\mathbb{E}_\rho[\mu(x)]$.

Random preparations also obey the usual law of total expectation. If state ρ_i is prepared with probability p_i , then the average state is $\rho = \sum_i p_i \rho_i$ and

$$\mathbb{E}_\rho[O] = \text{tr} \left[\left(\sum_i p_i \rho_i \right) O \right] = \sum_i p_i \mathbb{E}_{\rho_i}[O].$$

For $O = \mu(x)$, the same identity is precisely the law of total probability for the measurement outcome x .

Exercise 1.28 (An expectation value is an average). For $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13, calculate the expectation value of Z using both the outcome probabilities and the trace formula.

Solution. The outcomes $+1$ and -1 correspond to $|0\rangle$ and $|1\rangle$, respectively. Their probabilities are $3/4$ and $1/4$, so

$$\mathbb{E}(\text{outcome}) = (+1)\frac{3}{4} + (-1)\frac{1}{4} = \frac{1}{2}.$$

The trace formula gives the same result:

$$\text{tr}[Z\rho] = \text{tr} \begin{pmatrix} 3/4 & 1/4 \\ -1/4 & -1/4 \end{pmatrix} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

A single measurement returns $+1$ or -1 . The expectation value $1/2$ describes the average over many repetitions on identically prepared systems. $\square$